But, $\sec y = -2 \Rightarrow \cos y = -\frac{1}{2}$.

Now, $\cos y = -\frac{1}{2} = -\cos\frac{\pi}{3} = \cos\left(\pi - \frac{\pi}{3}\right) = \cos\frac{2\pi}{3}$. Therefore, $y = \frac{2\pi}{3}$.

Since $\frac{2\pi}{3} \in [0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$, the principal value of $\sec^{-1}(-2)$ is $\frac{2\pi}{3}$.

Example 4.13

Find the value of $\sec^{-1}\left(-\frac{2\sqrt{3}}{3}\right)$.

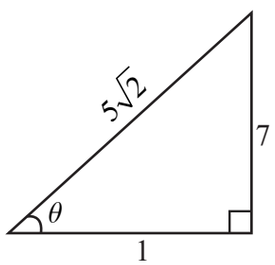
Solution

Let $\sec^{-1}\left(-\frac{2\sqrt{3}}{3}\right) = \theta$. Then, $\sec\theta = -\frac{2}{\sqrt{3}}$ where $\theta \in [0,\pi] \setminus \left\{\frac{\pi}{2}\right\}$. Thus, $\cos\theta = -\frac{\sqrt{3}}{2}$.

Now, $\cos\frac{5\pi}{6} = \cos\left(\pi - \frac{\pi}{6}\right) = -\cos\left(\frac{\pi}{6}\right) = -\frac{\sqrt{3}}{2}$. Hence, $\sec^{-1}\left(-\frac{2\sqrt{3}}{3}\right) = \frac{5\pi}{6}$.

Example 4.14

If $\cot^{-1}\left(\frac{1}{7}\right) = \theta$, find the value of $\cos\theta$.



Solution

By definition, $\cot^{-1} x \in (0,\pi)$.

Therefore, $\cot^{-1}\left(\frac{1}{7}\right) = \theta$ implies $\theta \in (0,\pi)$.

But $\cot^{-1}\left(\frac{1}{7}\right) = \theta$ implies $\cot\theta = \frac{1}{7}$ and hence $\tan\theta = 7$ and θ is acute.

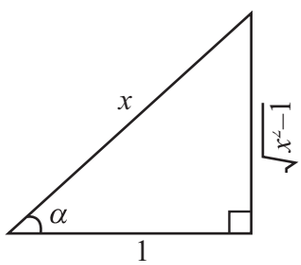
Using $\tan\theta = \frac{7}{1}$, we construct a right triangle as shown. Then, we have, $\cos\theta = \frac{1}{5\sqrt{2}}$.

Example 4.15

Show that $\cot^{-1}\left(\frac{1}{\sqrt{x^2-1}}\right) = \sec^{-1} x$, $|x| > 1$.

Solution

Let $\cot^{-1}\left(\frac{1}{\sqrt{x^2-1}}\right) = \alpha$. Then, $\cot\alpha = \frac{1}{\sqrt{x^2-1}}$ and α is acute.



We construct a right triangle with the given data.

From the triangle, $\sec\alpha = \frac{x}{1} = x$. Thus, $\alpha = \sec^{-1} x$.

Hence, $\cot^{-1}\left(\frac{1}{\sqrt{x^2-1}}\right) = \sec^{-1} x$, $|x| > 1$.

EXERCISE 4.4

1. Find the principal value of

(i) $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (ii) $\cot^{-1}\left(\sqrt{3}\right)$ (iii) $\operatorname{cosec}^{-1}\left(-\sqrt{2}\right)$

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